

## Dialogue Systems Lab 3 Task 3

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The system has multiple limitations. For example, while the speech recognizer could transcribe longer phrases, the dialogue manager will only accept exact grammar matches, which limits natural interaction. The app can handle certain phrases like “no way” and single-word inputs that are explicitly defined in the grammar, but not arbitrary longer sentences. Secondly, it is sensitive to speech recognition errors. If the words are slightly different or pronounced differently, the app might fail to recognize the input and repeat the same question. The same thing could happen in a noisy environment. I also noticed this happening when I tried to give my input louder, and I am not sure if this has to do with pronunciation change, how the device perceives louder noises or if it is strictly an app limitation. In addition, because the system directly correlates each input to one entity, it does not handle ambiguity or multiple interpretations. Also, casual flexible time expressions like “half past 10” and “in the afternoon” are not recognized by the app. Finally, something that is not exactly a limitation but could be changed to achieve a more natural interaction is how the system will tell you that your appointment is at thirteen hundred hours instead of 1 pm. This is technically a correct formal format, but it could be adjusted to something more informal and casual.

To help the system’s flow, I implemented retry states for every step, including states asking the user to give a yes/no answer. I did it in every step and tried to phrase it as clearly as possible so that the system doesn’t just stop or get stuck when it doesn’t understand something. I tried to make the retry states into something that is not just a repetition of the original question so that the interaction is more natural and the app doesn’t sound broken to the user. I also tried to extend the grammar with several appropriate yes/no variations to reduce the possibility of recognition failure and again, make the interaction more natural. I also adjusted the speech recognition timeout (`asrDefaultCompleteTimeout`) to prevent the system from staying in the listening state for too long, and in the `spst.listen` action I added an empty value object to the `LISTEN` message sent to the speech state actor so that I could prevent the app from getting stuck during recognition because of errors about missing timeout settings.